

SIDDHANTA MOHANTY

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EDUCATION

The Pennsylvania State University

Bachelor of Science in Computer Science

University Park, PA

Aug. 2022 – May 2026

- **Dean's List:** Fall 2023, Spring 2024, Fall 2025, Spring 2026 **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Computer Organization, Theory of Computation, Deep Learning for Computer Vision

EXPERIENCE

Research Assistant

Penn State University

Sep. 2025 – Dec. 2025

University Park, PA

- Designed **Reinforcement Learning** approaches to improve two-LLM code verification pipelines through iterative prompt refinement and compiler feedback
- Designed integration strategies for combining **RL** with validation tools (**Klee**, **CodeQL**) to enable automated code evaluation, self-correction and self-learning

GenAI Intern

HCL Technologies

May 2025 – Aug. 2025

Redmond, WA

- Built a **RAG-based support automation system** using Flask, LangChain, and FAISS that analyzed production error logs, retrieved similar incidents and docs, and sent Slack alerts with remediation recommendations, reducing support resolution time by **50%**
- Designed **retrieval pipelines** using FAISS and structured metadata to index internal docs and historical incidents
- Orchestrated **multi-agent workflows** across APIs, databases, and Confluence using **MCP** and LangChain

Data Analytics Intern

Elevatoz Loyalty

June 2024 – Aug. 2024

Bengaluru, India

- Applied **SQL** and **Python (Pandas, NumPy)** to clean, transform, and analyze structured datasets.

PROJECTS

Self-Heal System | Flask, LangChain, FAISS, PostgreSQL, OpenAI, Slack

[Source Code](#)

- Built an **AI-powered self-healing backend** that embedded error logs into FAISS, retrieved similar past incidents via semantic search, and generated context-aware remediation recommendations.
- Developed a **Slack-integrated incident response system** that delivered severity-based alerts, LLM-generated diagnostics and fixes, and a chatbot with access to recent error logs, database statistics, and a read-only SQL interface restricted to **SELECT** queries.

Rewind | Python, Playwright, Pydantic, Claude API, MCP

[Source Code](#)

- Built an **LLM-driven browser agent** using Claude tool-calling and Playwright that completes multi-step workflows on a demo banking app from accessibility-tree snapshots and compiles successful runs into **typed, versioned replay artifacts**.
- Engineered a **deterministic replay engine** with **zero LLM calls** that runs recorded workflows (e.g., loan requests) in **0.41s vs 13.1s (32x faster)**, exposed to AI agents through an **MCP server** with allowlisting, approval gating, and secret redaction.

Prompt Tracer | JavaScript, Chrome Extension API, Manifest V3

[Source Code](#)

- Developed a **prompt observability Chrome extension** that analyzes and scores prompts in real time using a deterministic 5-point rubric, injecting a live feedback panel into ChatGPT, Claude, Grok, and Gemini for instant, consistent scoring and suggestions
- Built a dual-mode prompt optimization engine combining a privacy-safe rule-based optimizer with an optional OpenAI-powered rewrite pipeline, using ambiguity detection to distinguish prompts requiring clarification from those needing structural improvements

yourAlbrief | Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, Resend, Vercel

[Demo](#) | [Source Code](#)

- Built a full-stack **AI newsletter platform** with magic-link authentication, personalized topic preferences and delivery schedules, aggregating content from 50+ trusted sources and tracking per-issue email open and click analytics.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks/Libraries: Flask, FastAPI (REST APIs), Django, Next.js, Node.js, React, LangChain, SQLAlchemy, Pydantic

AI/ML Tools: RAG, LLMs, AI Agents, Tool Calling, Model Context Protocol (MCP), Claude API, OpenAI API, NumPy, PyTorch

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases (FAISS, Pinecone)

Developer Tools: Git, GitHub, GitHub Actions, Docker, Playwright, Prisma, Postman, AWS, Google Cloud Platform, VS Code, ROS2